



CS 540 Introduction to Artificial Intelligence

Unsupervised Learning I

Spring 2025

Announcements

- **Homework:**
 - HW4 to be released today
- Class roadmap:

ML Unsupervised I
ML Unsupervised II
ML Linear Regression
Machine Learning: K - Nearest Neighbors & Naive Bayes

Machine Learning

Recap of Supervised/Unsupervised

Supervised learning:

- Make predictions, classify data, perform regression
- Dataset: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \dots, (\mathbf{x}_n, y_n)$



Features / Covariates / Input

Labels / Outputs

- Goal: find function $f : X \rightarrow Y$ to predict label on **new** data



indoor

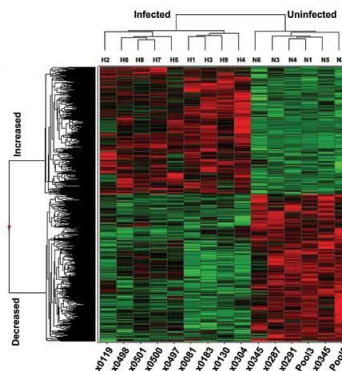
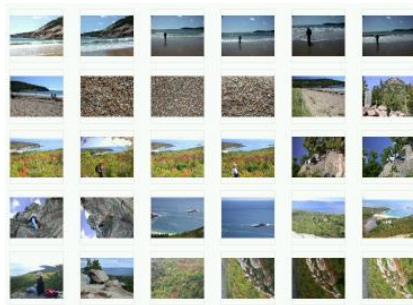
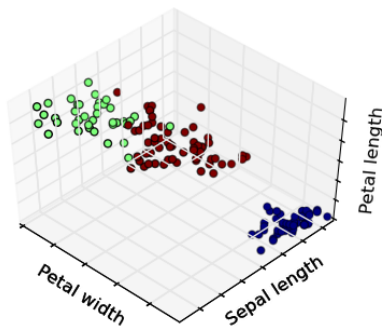


outdoor

Recap of Supervised/Unsupervised

Unsupervised learning:

- No labels; generally, won't be making predictions
- Dataset: $X_1, X_2, \dots, X_n$
- Goal: find patterns & structures that help better understand data.

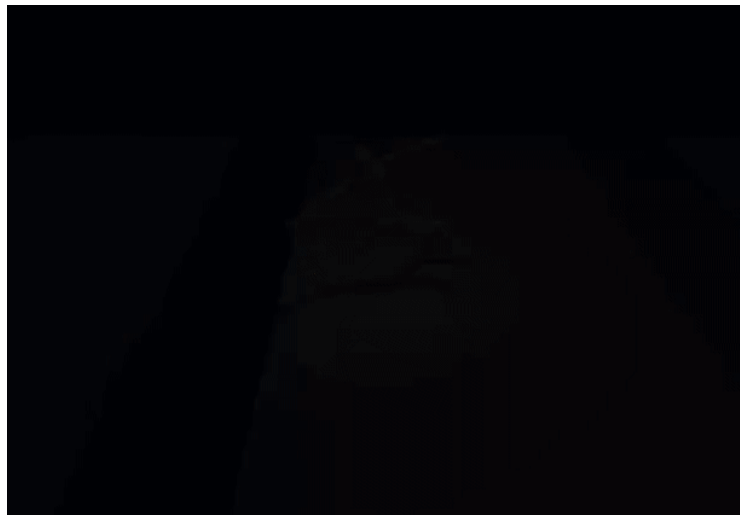


Mulvey and Gingold

Recap of Supervised/Unsupervised

Note that there are **other kinds** of ML:

- Mixtures: semi-supervised learning, self-supervised
 - Idea: different types of “signal”
- Reinforcement learning
 - Learn how to act in order to maximize rewards
 - Later on in course...



DeepMind

Outline

- Intro to Clustering
 - Clustering Types
- Hierarchical Clustering
 - Divisive, agglomerative, linkage strategies
- Centroid-based Clustering
 - k-means

Unsupervised Learning & Clustering

- Note that clustering is just one type of unsupervised learning (UL)
 - ★ – PCA is another unsupervised algorithm
- Estimating probability distributions also UL (GANs)
- Clustering is popular & useful!



StyleGAN2 (Kerras et al '20)

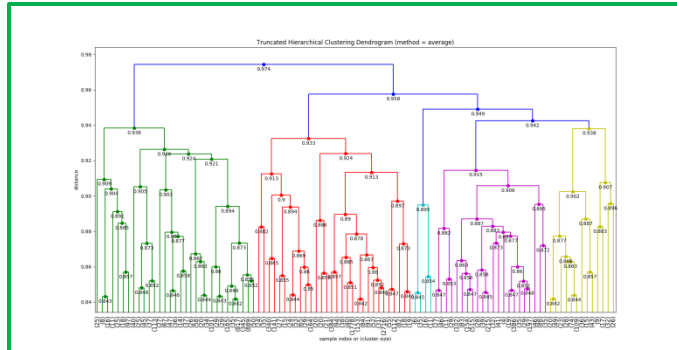
Clustering Types

- Several types of clustering



Hierarchical

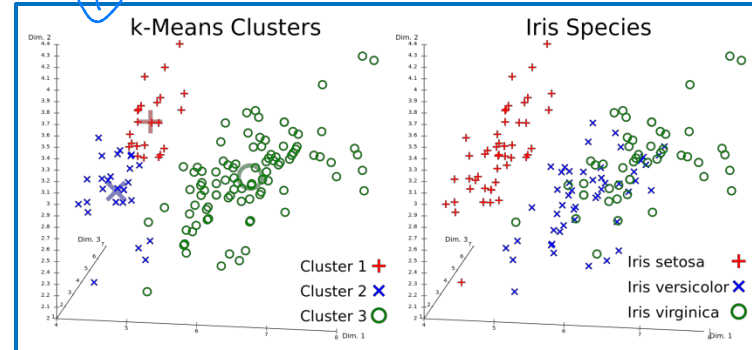
- Agglomerative
- Divisive



Partitional

- Center-based
- Graph-theoretic
- Spectral

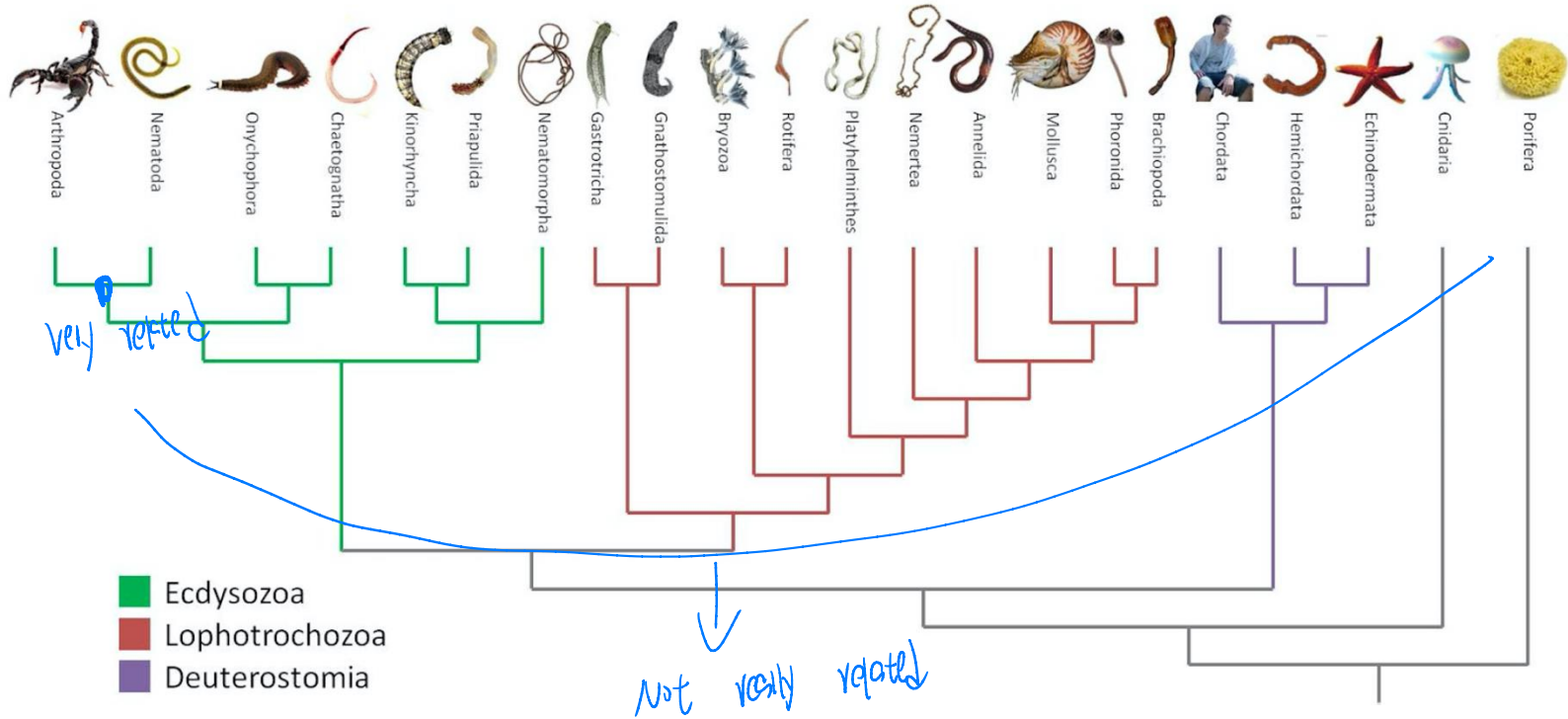
Assign points to some grouping



Bayesian

- Decision-based
- Nonparametric

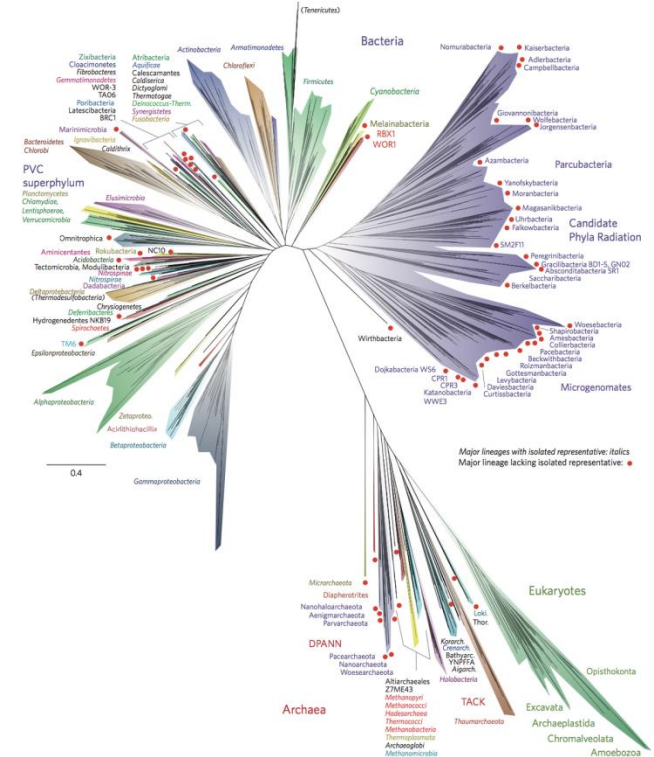
Hierarchical Clustering



Hierarchical Clustering

Basic idea: build a “hierarchy”

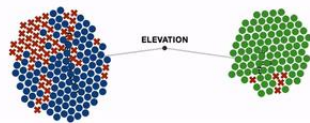
- Want: arrangements from specific to general
- ✧ One advantage: no need for k, number of clusters.
- ✧ **Input:** points. **Output:** a hierarchy
 - A binary tree



Agglomerative vs Divisive

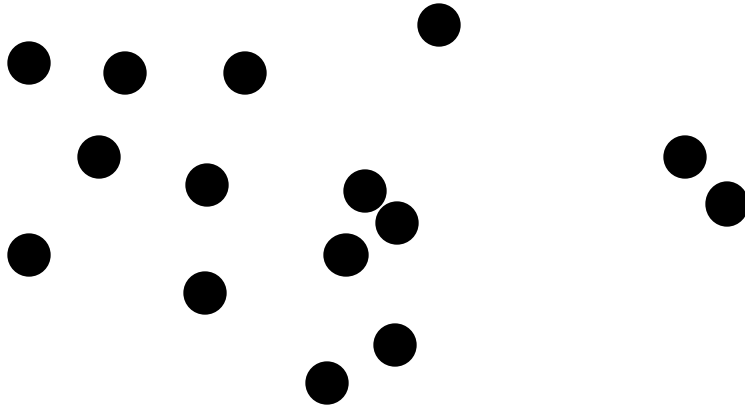
Two ways to go:

- ★ • **Agglomerative:** bottom up.
 - Start: each point a cluster. Progressively merge clusters (merge points to bigger clusters)
- ★ • **Divisive:** top down
 - Start: all points in one cluster. Progressively split clusters (split to smaller clusters)



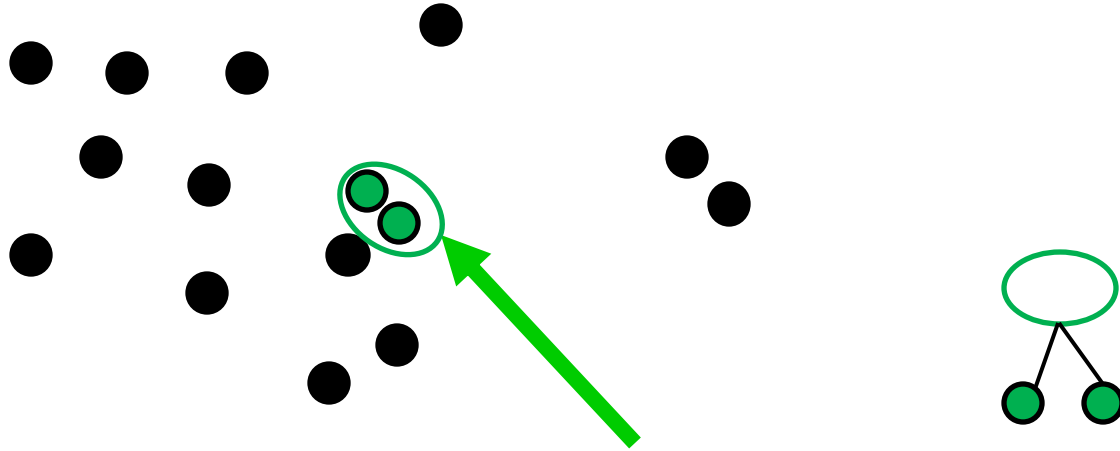
Agglomerative Clustering Example

Agglomerative. Start: every point is its own cluster



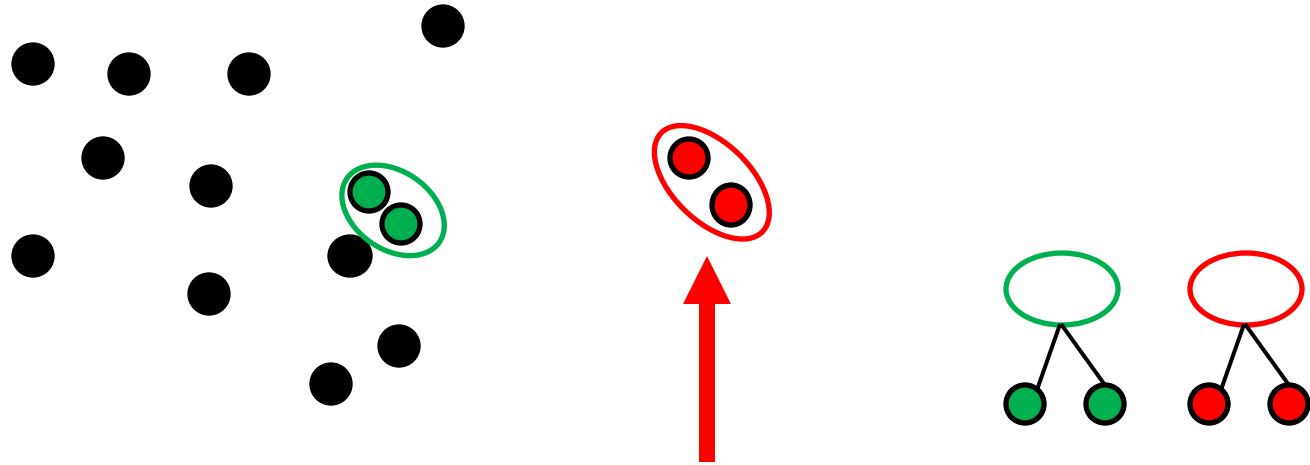
Agglomerative Clustering Example

Get pair of clusters that are closest and merge



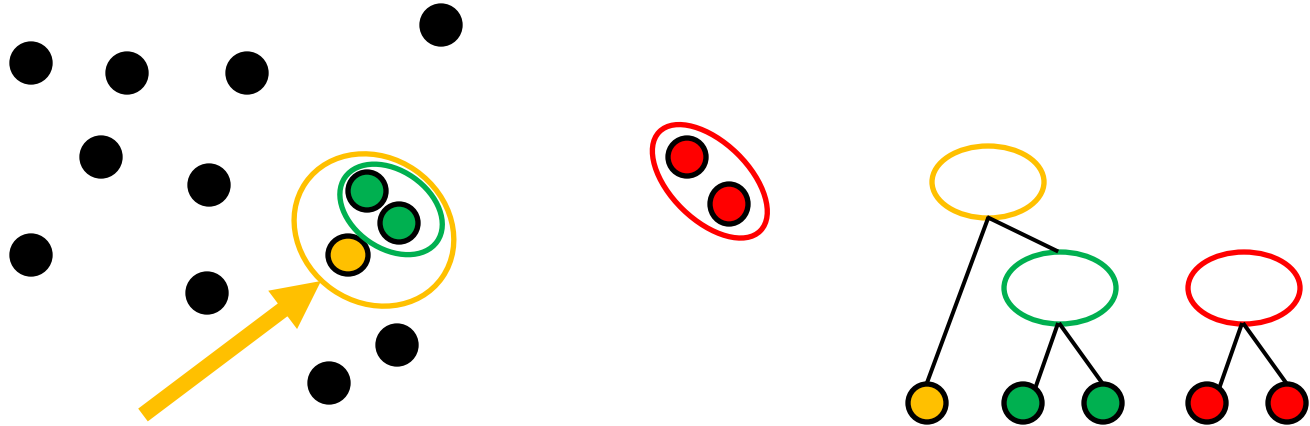
Agglomerative Clustering Example

Repeat: Get pair of clusters that are closest and merge



Agglomerative Clustering Example

Repeat: Get pair of clusters that are closest and merge



Merging Criteria

Merge: use closest clusters. Define closest?

- Single-linkage ↗ A, B here are sets of points

$$d(A, B) = \min_{x_1 \in A, x_2 \in B} d(x_1, x_2)$$

↘ smallest possible distance between 2 sets



- Complete-linkage

$$d(A, B) = \max_{x_1 \in A, x_2 \in B} d(x_1, x_2)$$

↘ further possible distance between 2 sets



- Average-linkage

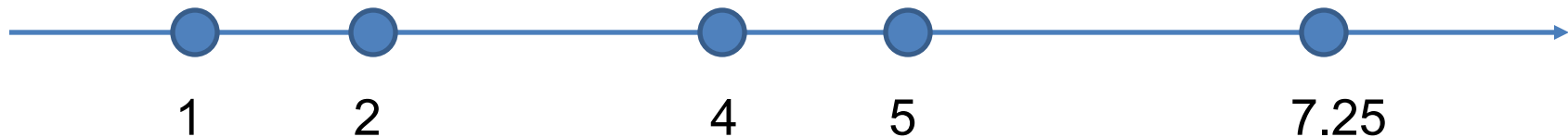
$$d(A, B) = \frac{1}{|A||B|} \sum_{x_1 \in A, x_2 \in B} d(x_1, x_2)$$

↙ size of A and B multiplication

Single-linkage Example

We'll merge using single-linkage

- 1-dimensional vectors.
- Initial: all points are clusters

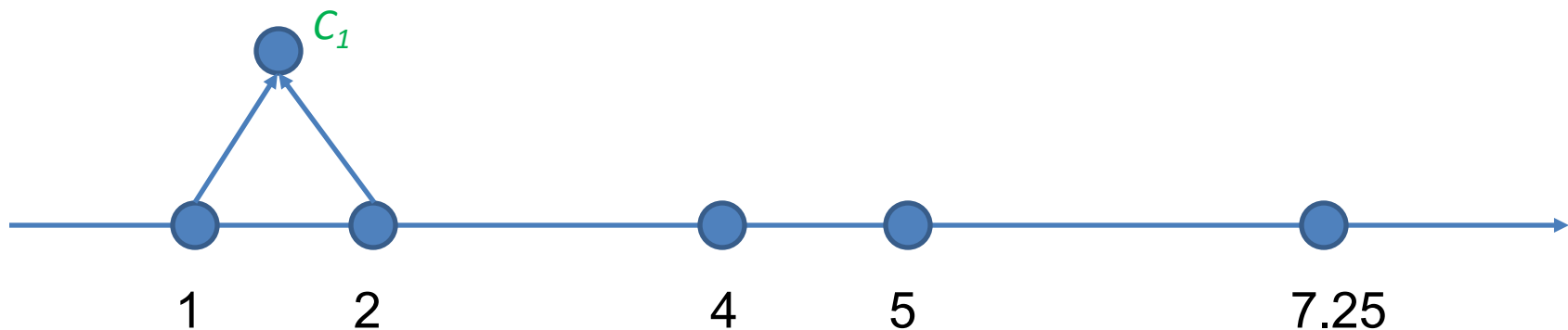


Single-linkage Example

We'll merge using single-linkage

$$d(C_1, \{4\}) = d(2, 4) = 2$$

$$d(\{4\}, \{5\}) = d(4, 5) = 1$$

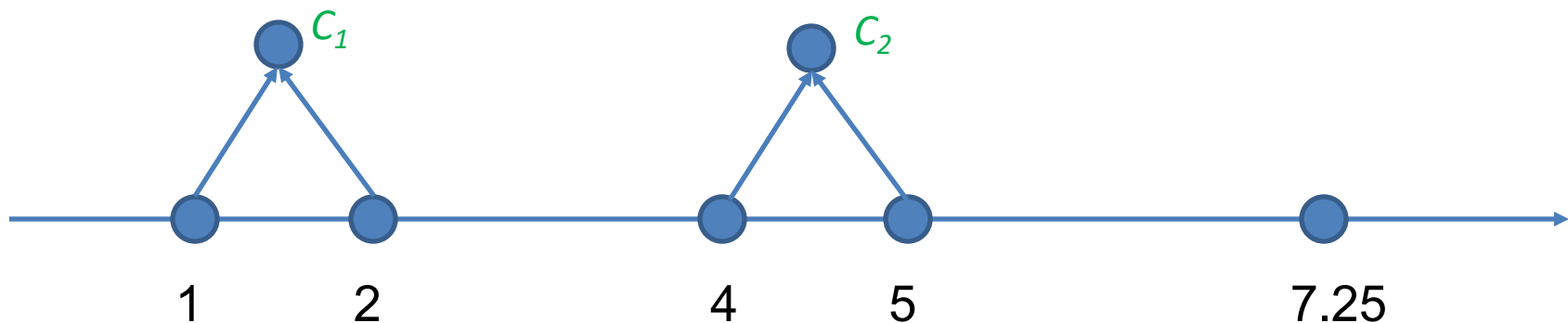


Single-linkage Example

Continue...

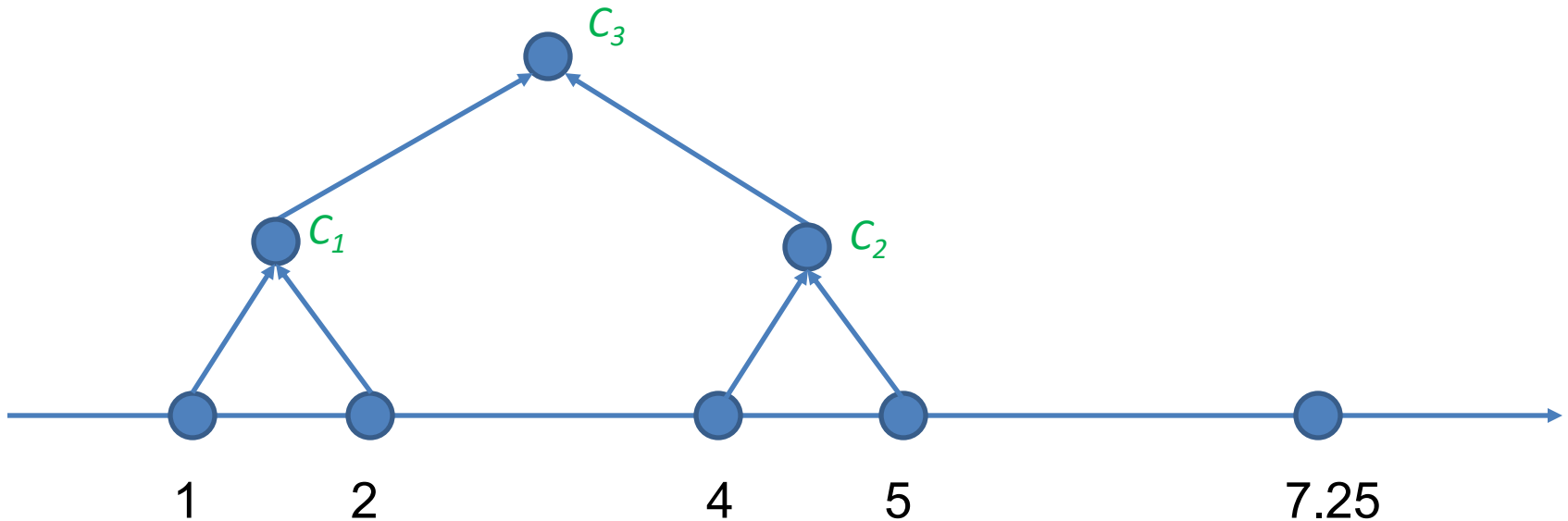
$$d(C_1, C_2) = d(2, 4) = 2$$

$$d(C_2, \{7.25\}) = d(5, 7.25) = 2.25$$

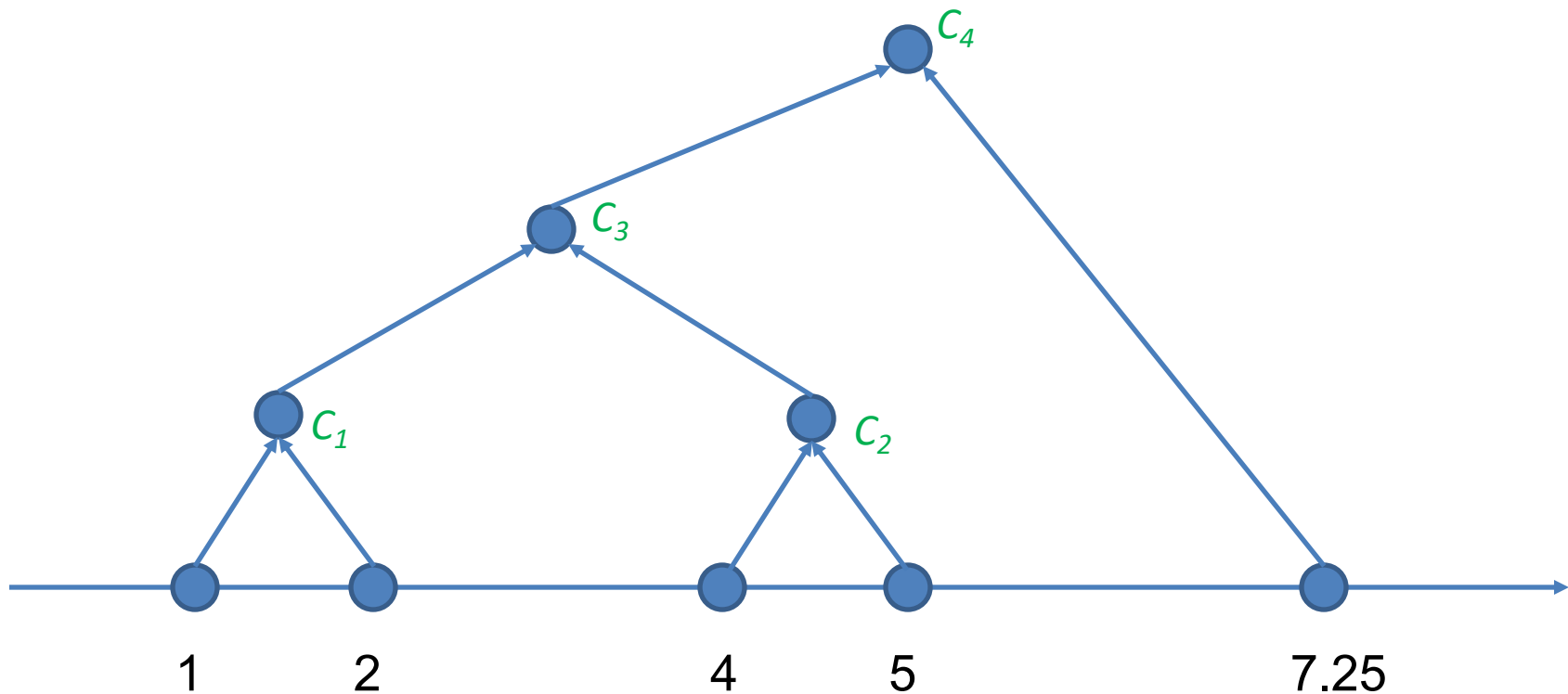


Single-linkage Example

Continue...



Single-linkage Example

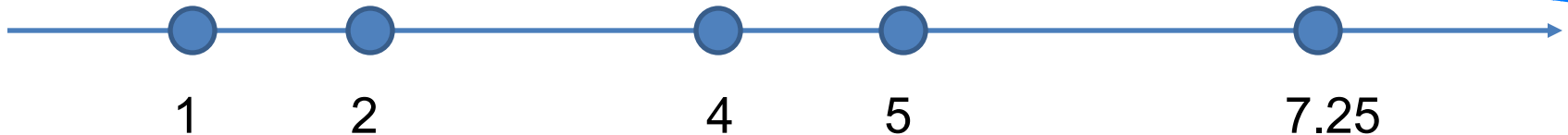


Complete-linkage Example

We'll merge using complete-linkage

- 1-dimensional vectors.
- Initial: all points are clusters

★ Very important: we are still merging the clusters for min distance, it's just the way of calculation of the distance between two clusters change!!! The distance between each pair of cluster is $\max d(x_i, x_j)$, but we find the minimum of all combinations!

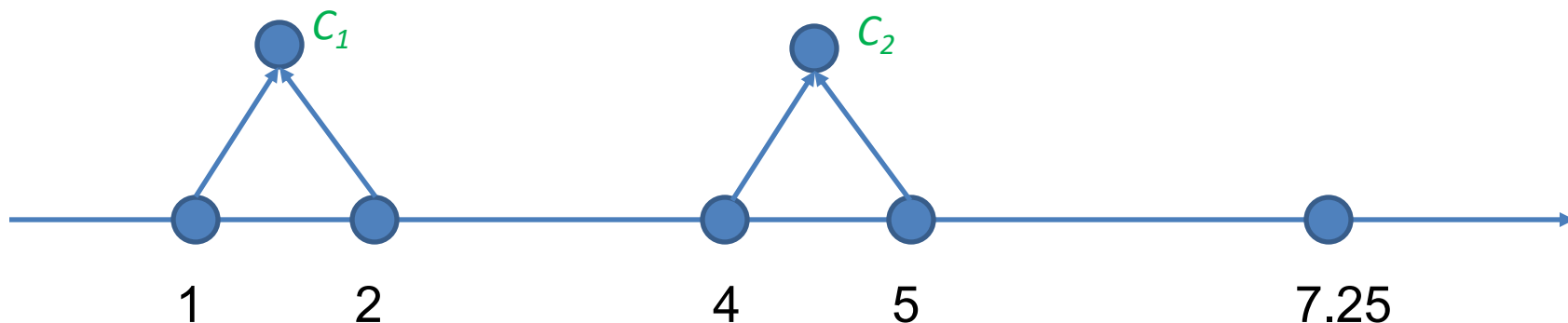


Complete-linkage Example

Beginning is the same...

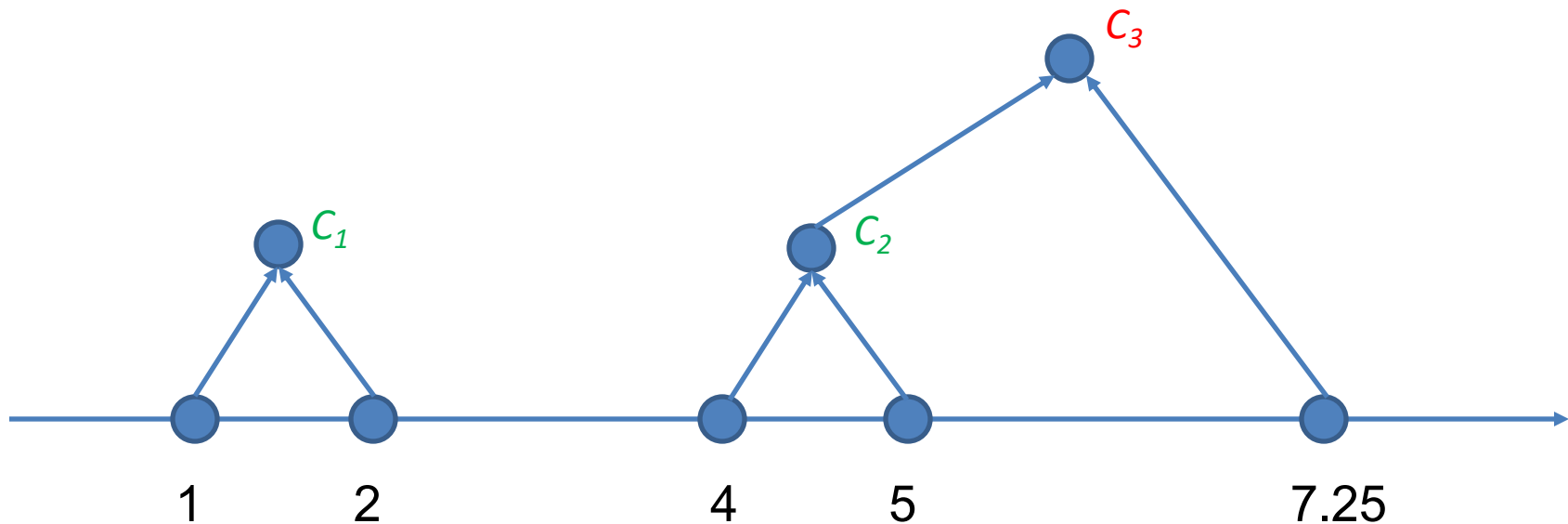
$$d(C_1, C_2) = d(1, 5) = 4$$

$$d(C_2, \{7.25\}) = d(4, 7.25) = 3.25$$

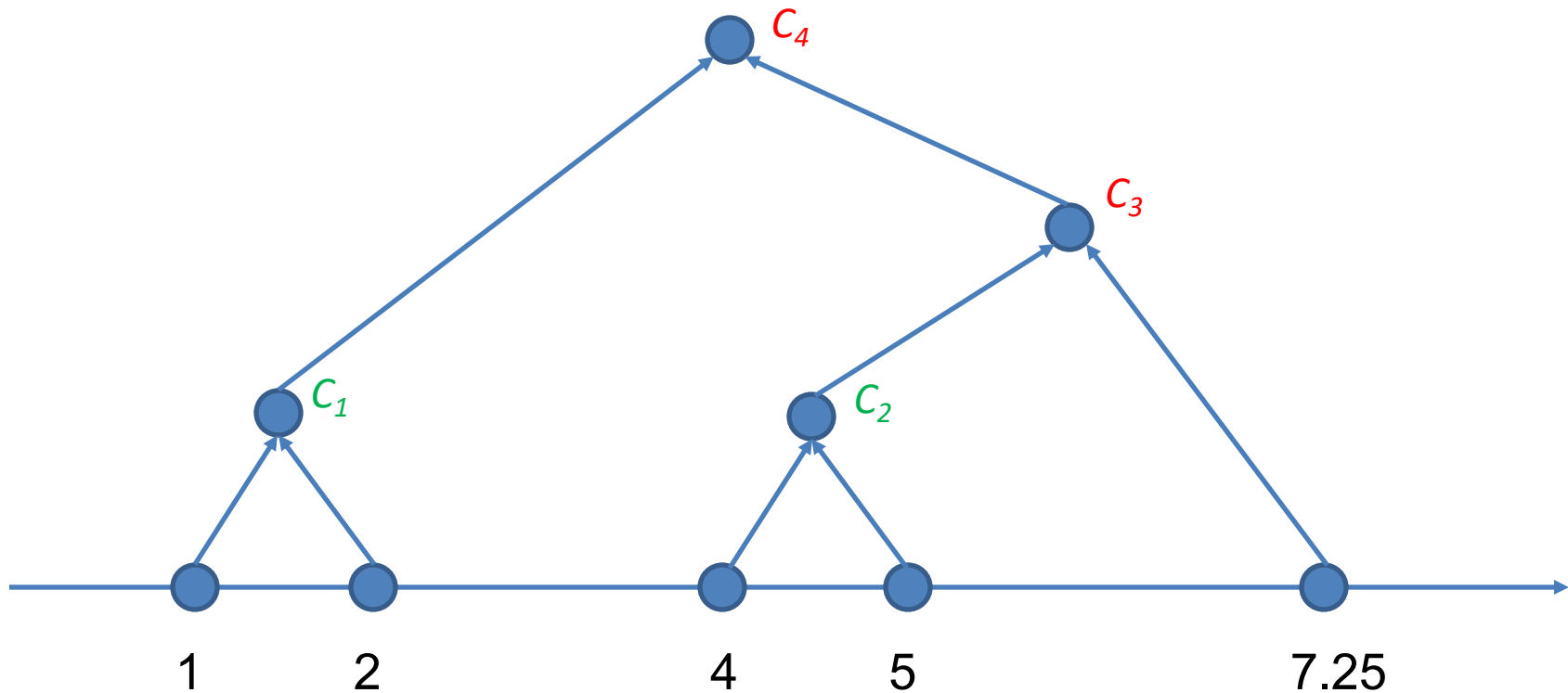


Complete-linkage Example

Now we diverge:



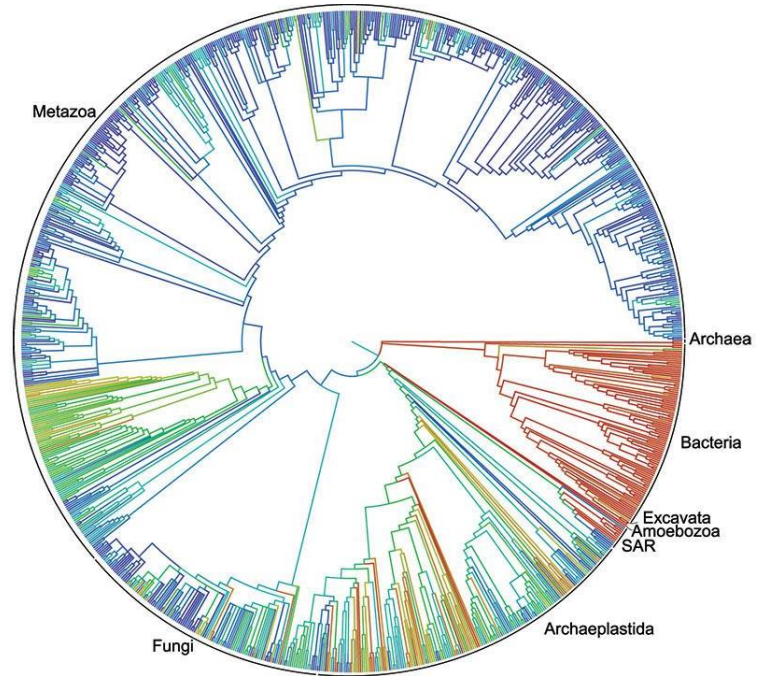
Complete-linkage Example



When to Stop?

No simple answer:

- Use the binary tree (a **dendrogram**)
- Cut at different levels (get different heights/depths)

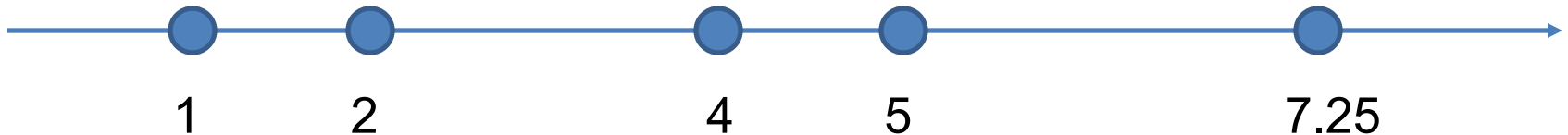


<http://opentreeoflife.org/>

Break & Quiz

Q 1.1: Let's do hierarchical clustering for two clusters with average linkage on the dataset below. What are the clusters?

- A. $\{1\}, \{2, 4, 5, 7.25\}$
- B. $\{1, 2\}, \{4, 5, 7.25\}$
- C. $\{1, 2, 4\}, \{5, 7.25\}$
- D. $\{1, 2, 4, 5\}, \{7.25\}$



Break & Quiz

Q 1.1: Let's do hierarchical clustering for two clusters with average linkage on the dataset below. What are the clusters?

- A. {1}, {2,4,5,7.25}
- **B. {1,2}, {4, 5, 7.25}**
- C. {1,2,4}, {5, 7.25}
- D. {1,2,4,5}, {7.25}

Iteration 1: merge 1 and 2

Iteration 2: merge 4 and 5

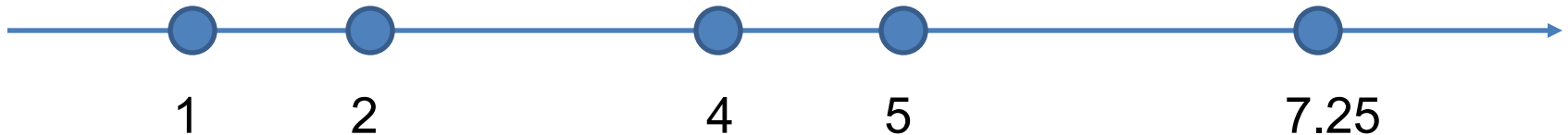
Iteration 3: Now we have clusters {1,2}, {4,5}, {7.25}.

$\text{distance}(\{1,2\}, \{4,5\}) = 3$

$\text{distance}(\{4,5\}, \{7.25\}) = 2.75$

$\text{distance}(\{1,2\}, \{7.25\})$ is clearly larger than the above two.

So average linkage will merge {4,5} and {7.25}



Break & Quiz

Q 1.2: If we do hierarchical clustering on n points, the maximum depth of the resulting tree is

- A. 2
- B. $\log n$
- C. $n/2$
- D. $n-1$

Break & Quiz

Q 1.2: If we do hierarchical clustering on n points, the maximum depth of the resulting tree is

- A. 2
- B. $\log n$
- C. $n/2$
- **D. $n-1$**

Denote the points as $x_1, x_2, \dots, x_n$

Suppose:

in iteration 1, we merge points x_1 and x_2

in iteration 2, we merge $\{x_1, x_2\}$ with x_3

...

in iteration t , we merge $\{x_1, x_2, \dots, x_t\}$ with x_{t+1}

...

in iteration $n-1$, we merge $\{x_1, x_{n-1}\}$ with x_n

Then we will get a tree with depth $n-1$.

Center-based Clustering

- ★ • k-means is an example of a partitional, center-based clustering algorithm.
- Specify a desired number of clusters, k ; run k-means to find k clusters.

K-means clustering

- Very popular clustering method

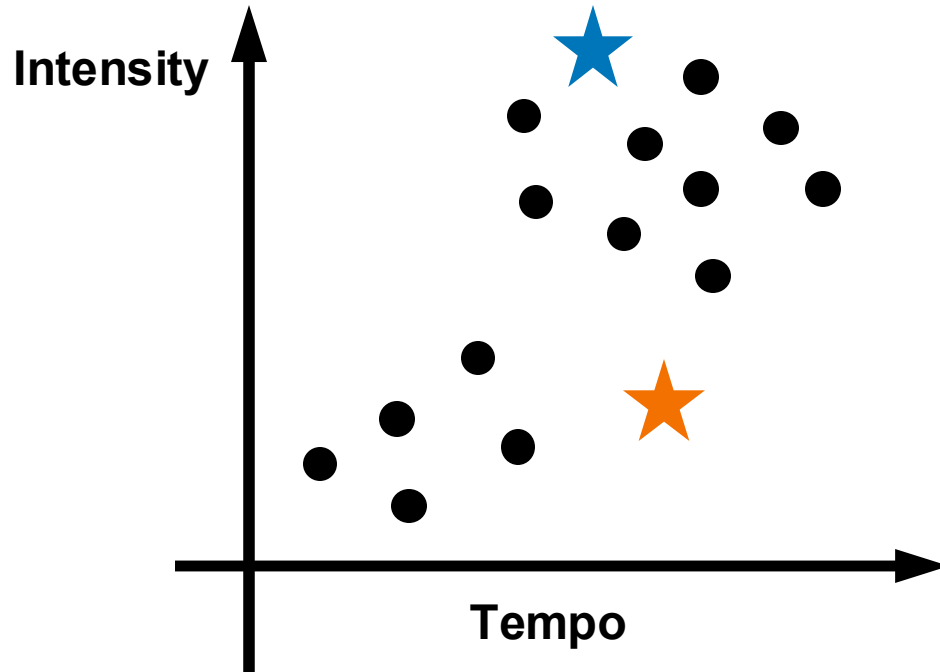
$$x_1, x_2, \dots, x_n$$

Input

- Input: a dataset, and assume the number of clusters **k** is given

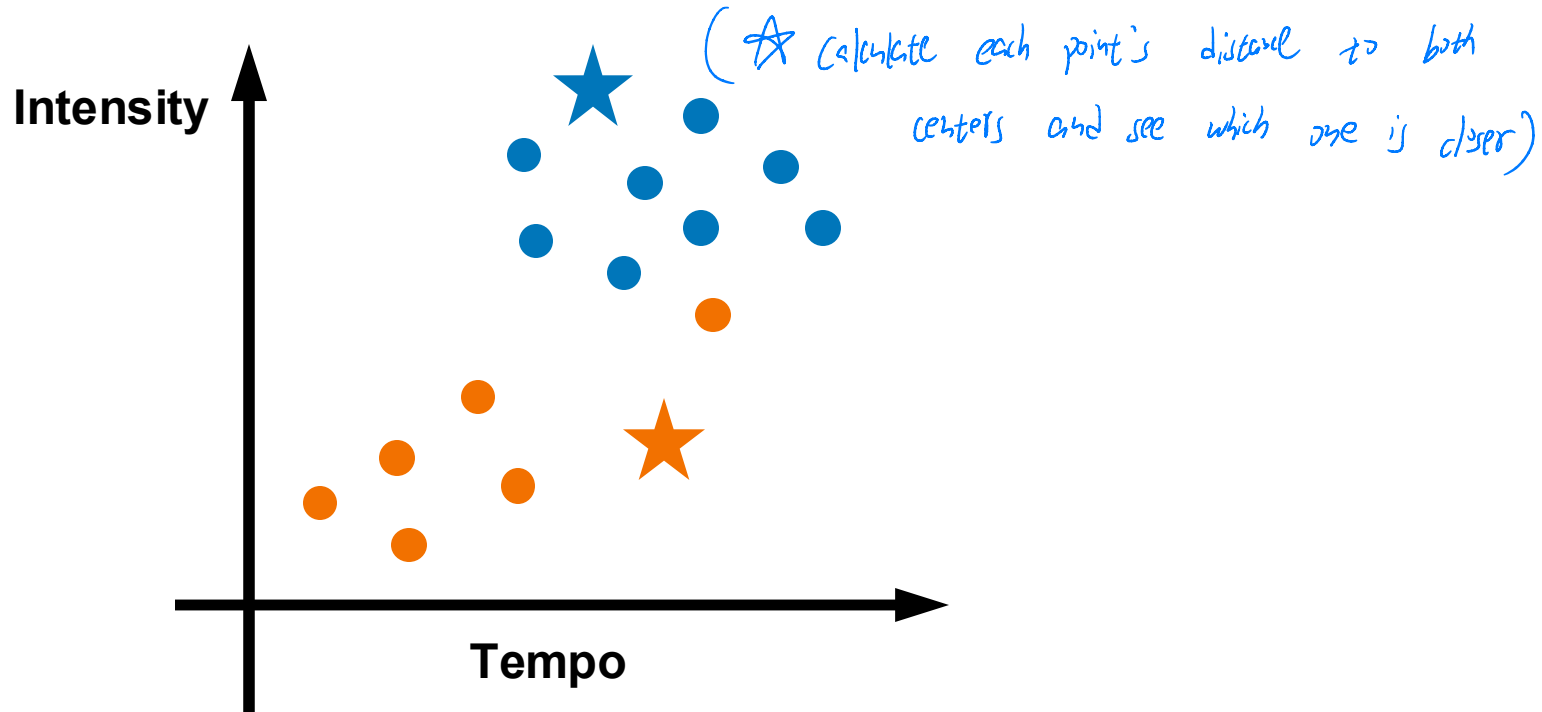
K-means clustering

Step 1: **Randomly** picking 2 positions as initial cluster centers
(not necessarily a data point)



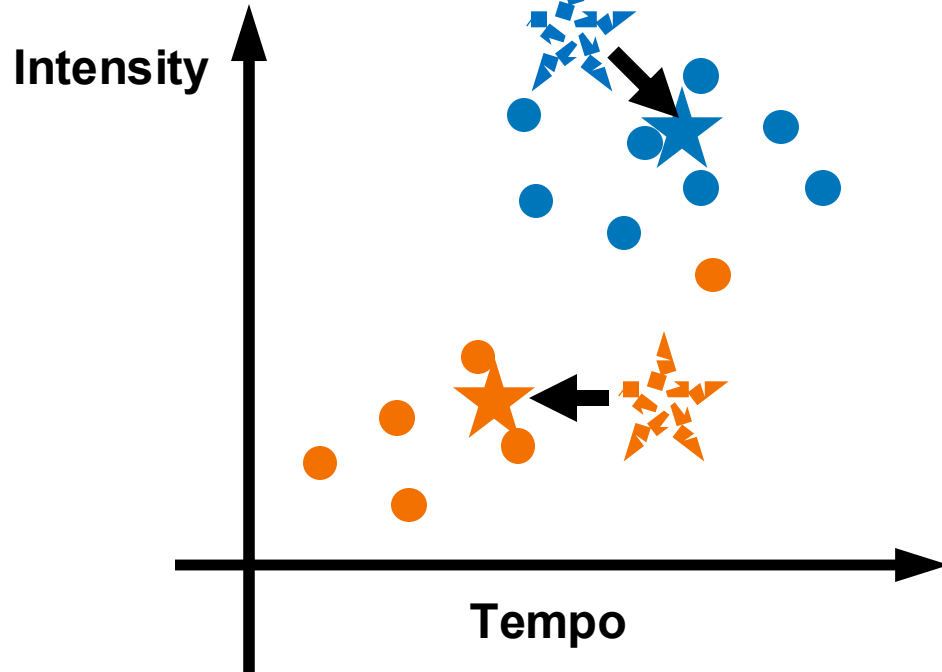
K-means clustering

Step 2: for each point x , determine its cluster: find the closest center in Euclidean space



K-means clustering

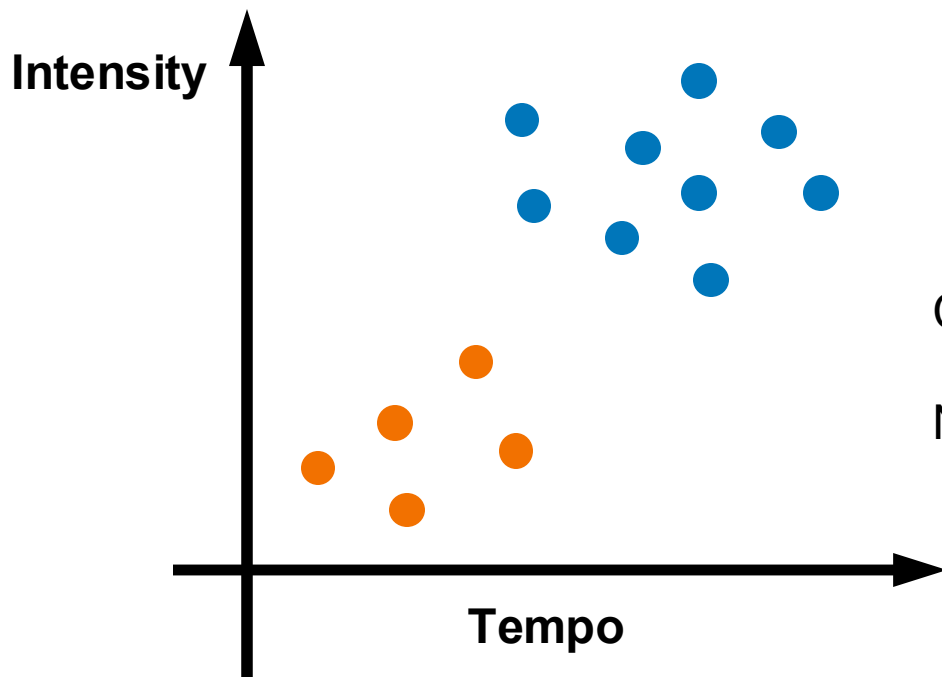
Step 3: update all cluster centers as the centroids



↓
(★ Take the average of all data points in the previous cluster and shift the previous random center to a new center)

K-means clustering

Repeat step 2 & 3 until convergence

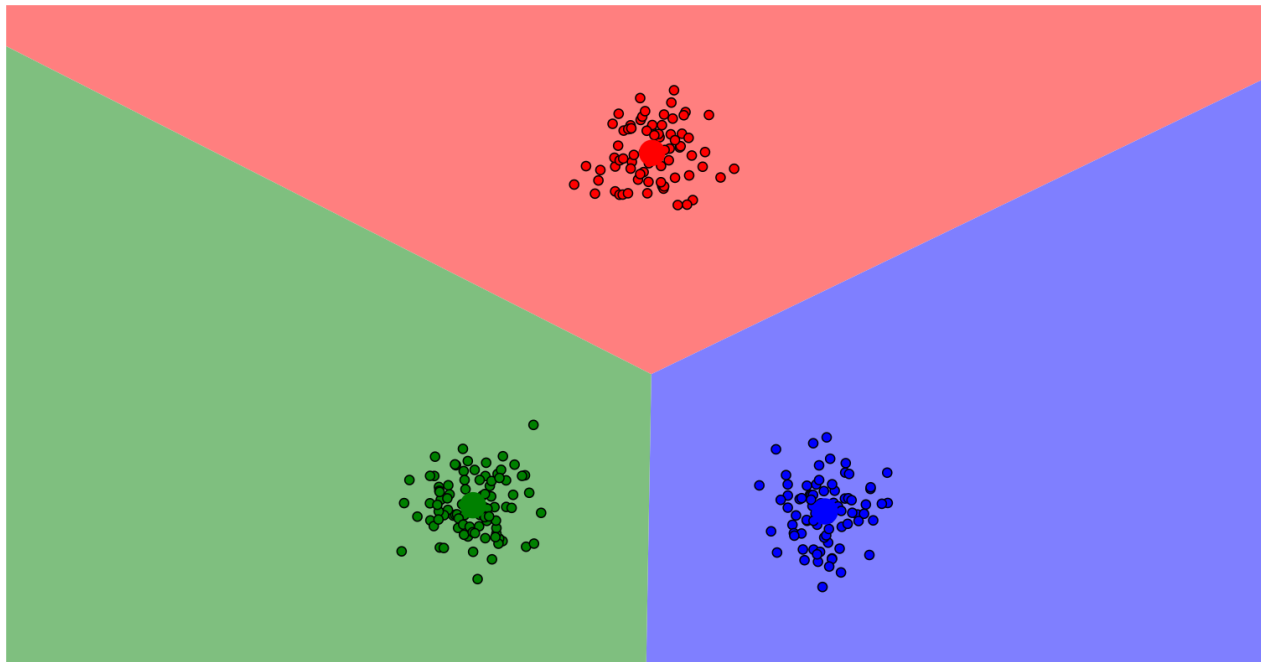


Converged solution!

No labels required!

K-means clustering: A demo

<https://www.naftaliharris.com/blog/visualizing-k-means-clustering/>



★ K-means algorithm

- Input: $x_1, x_2, \dots, x_n, k$
- Step 1: ^{Randomly} select k cluster centers $c_1, c_2, \dots, c_k$
- Step 2: for each point x_i , assign it to the closest center in Euclidean distance:

Which cluster is x_i in $\leftarrow y(x_i) = \operatorname{argmin}_j \|x_i - c_j\|$ $\rightarrow j \text{ from } 1 \text{ to } k$

- Step 3: update all cluster centers as the centroids:

$$c_j = \frac{\sum_{x: y(x)=j} x}{\sum_{x: y(x)=j} 1}$$

★ $\frac{1}{|S_j|} \sum_{x_i \in S_j} x_i$

$|S_j|$ is the num of points in cluster j

★ The output is just $y(x_i)$

- Repeat Step 2 and 3 until cluster centers no longer change

Questions on k-means

- What is k-means trying to optimize?
- Will k-means stop (converge)?
- Will it find a global or local optimum?
- How to pick starting cluster centers?
- How many clusters should we use?

The optimization problem of k-means

- What is k-means trying to optimize?

$$\min_{c,y} \sum_{i=1}^n ||x_i - c_{y(x_i)}||^2$$

Questions on k-means

Will k-means stop (converge)? Yes

Given a fixed dataset and a fixed number of clusters, there are only a **finite number of ways** to assign data points to clusters.

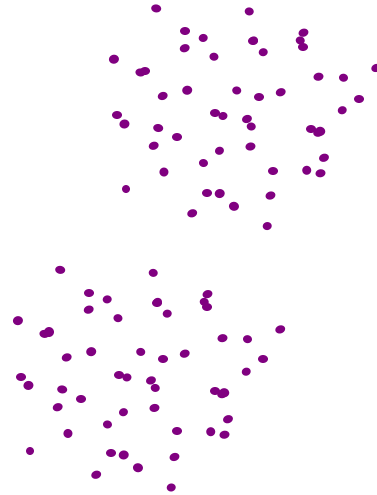
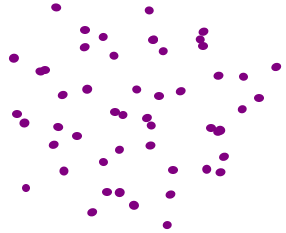
Each iteration consists of:

- Assignment Step: Assign each data point to the closest centroid.
- Update Step: Recompute centroids as the mean of assigned points.

These steps **always reduce or keep the same** the objective function (sum of squared distance), ensuring termination.

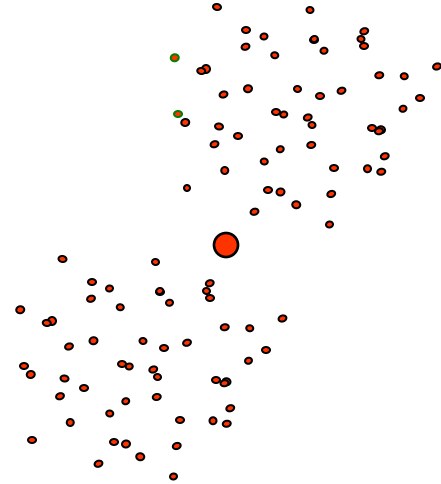
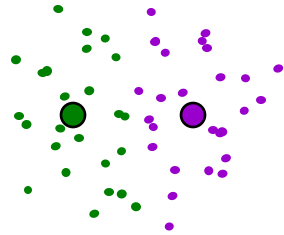
Questions on k-means

Will it find a global or local optimum? (sadly no guarantee)



Questions on k-means

Will it find a global or local optimum? (sadly no guarantee)



Questions on k-means

How many clusters should we use?

- Difficult problem
- Domain knowledge
- Elbow Method
 - Compute the within-cluster sum of squares (WCSS) for different values of k .
 - Plot WCSS vs. k and look for the **elbow point** where the reduction in WCSS slows down. The optimal k is typically at this elbow.

Break & Quiz

Q 2.1: You have seven 2-dimensional points. You run 3-means on it, with initial clusters

$$C_1 = \{(2, 2), (4, 4), (6, 6)\}, C_2 = \{(0, 4), (4, 0)\}, C_3 = \{(5, 5), (9, 9)\}$$

Cluster centroids are updated to?

- A. $C_1: (4, 4)$, $C_2: (2, 2)$, $C_3: (7, 7)$
- B. $C_1: (6, 6)$, $C_2: (4, 4)$, $C_3: (9, 9)$
- C. $C_1: (2, 2)$, $C_2: (0, 0)$, $C_3: (5, 5)$
- D. $C_1: (2, 6)$, $C_2: (0, 4)$, $C_3: (5, 9)$

Break & Quiz

Q 2.1: You have seven 2-dimensional points. You run 3-means on it, with initial clusters

$$C_1 = \{(2, 2), (4, 4), (6, 6)\}, C_2 = \{(0, 4), (4, 0)\}, C_3 = \{(5, 5), (9, 9)\}$$

Cluster centroids are updated to?

- **A. $C_1: (4,4)$, $C_2: (2,2)$, $C_3: (7,7)$**
- B. $C_1: (6,6)$, $C_2: (4,4)$, $C_3: (9,9)$
- C. $C_1: (2,2)$, $C_2: (0,0)$, $C_3: (5,5)$
- D. $C_1: (2,6)$, $C_2: (0,4)$, $C_3: (5,9)$

The average of points in C1 is (4,4).

The average of points in C2 is (2,2).

The average of points in C3 is (7,7).

Break & Quiz

Q 2.2: We are running 3-means again. We have 3 centers, C_1 (0,1), C_2 (2,1), C_3 (-1,2). Which cluster assignment is possible for the points (1,1) and (-1,1), respectively? Ties are broken arbitrarily:

(i) C_1, C_1 (ii) C_2, C_3 (iii) C_1, C_3

- A. Only (i)
- B. Only (ii) and (iii)
- C. Only (i) and (iii)
- D. All of them

Break & Quiz

Q 2.2: We are running 3-means again. We have 3 centers, C_1 (0,1), C_2 (2,1), C_3 (-1,2). Which cluster assignment is possible for the points (1,1) and (-1,1), respectively? Ties are broken arbitrarily:

(i) C_1 , C_1 (ii) C_2 , C_3 (iii) C_1 , C_3

- A. Only (i)
- B. Only (ii) and (iii)
- C. Only (i) and (iii)
- **D. All of them**

For the point (1,1): square-Euclidean-distance to C_1 is 1, to C_2 is 1, to C_3 is 5

So it can be assigned to C_1 or C_2

For the point (-1,1): square-Euclidean-distance to C_1 is 1, to C_2 is 9, to C_3 is 1

So it can be assigned to C_1 or C_3

Break & Quiz

Q 2.3: If we run K-means clustering twice with random starting cluster centers, are we guaranteed to get same clustering results? Does K-means always converge?

- A. Yes, Yes
- B. No, Yes
- C. Yes, No
- D. No, No

Break & Quiz

Q 2.3: If we run K-means clustering twice with random starting cluster centers, are we guaranteed to get same clustering results? Does K-means always converge?

The clustering from k-means will depend on the initialization. Different initialization can lead to different outcomes.

- A. Yes, Yes
- **B. No, Yes**
- C. Yes, No
- D. No, No

K-means will always converge on a finite set of data points:

1. There are finite number of possible partitions of the points
2. The assignment and update steps of each iteration will only decrease the sum of the distances from points to their corresponding centers.
3. If it run forever without convergence, it will revisit the same partition, which is contradictory to item 2.



Thanks!